Business Intelligence and Predictive Analytics for Sales and Collections Performance Analysis

Done By:

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Abstract

This project investigates the use of data analytics, business intelligence, and machine learning techniques to analyze sales performance and efficiency of collection within an organization. Using real transactional and collections datasets, the study aims to support data-driven decision-making by uncovering patterns related to sales amounts, quantities, customer behavior, payment methods, and collection performance. The project integrates multiple analytical approaches to provide a comprehensive understanding of business operations and performance drivers.

The implementation phase involved data cleaning, transformation, and descriptive statistical analysis using Microsoft Excel and Python. Interactive dashboards were developed in Power BI to visualize sales trends by time, location, customer, product, and salesperson. Predictive analysis was conducted using RapidMiner, where Linear Regression and Random Forest models were built to predict sales amounts based on historical data. Model performance was evaluated using regression metrics such as correlation, R^2, and RMSE. In addition, K-Means clustering was applied to segment customers according to sales value and behavioral patterns.

The results demonstrate that the predictive models achieved strong performance, with the Random Forest model outperforming Linear Regression in terms of accuracy and error reduction. Customer clustering revealed distinct segments, including high-value customers, discount-sensitive customers, and low-priority segments, enabling targeted sales and collection strategies. Overall, the integration of descriptive analytics, predictive modeling, and clustering within a unified business intelligence framework proved effective in delivering actionable insights, enhancing sales forecasting, optimizing collections strategies, and improving overall financial performance.

Business Intelligence Project description and objectives

Project Description and Goal

This project focuses on applying Business Intelligence and data analytics techniques within the sales and distribution industry to improve decision-making related to sales performance and collections efficiency. Using real transactional and collections data, the project aims to transform raw operational data into meaningful insights through descriptive, predictive, and exploratory analysis. By integrating tools such as Microsoft Excel, RapidMiner, and Power BI, the project demonstrates how data-driven approaches can support organizations in understanding customer behavior, optimizing sales strategies, improving payment performance, and enhancing overall financial outcomes.

Project Objectives

Analyze Sales and Collections Performance:

Examine historical sales and collections data to identify trends, patterns, and performance gaps across customers, products, sales representatives, locations, and time periods using descriptive analytics and statistical summaries.

Predict Sales Amounts:

Develop and evaluate supervised machine learning models, including Linear Regression and Random Forest, to predict sales amounts based on key transactional, product, and customer-related factors.

Identify Key Business Drivers:

Determine the most influential variables affecting sales performance and collection efficiency, such as quantity sold, discounts, value category, customer behavior, and payment methods.

Segment Customers Using Clustering:

Apply unsupervised learning techniques, specifically K-Means clustering, to group customers according to their sales behavior and value characteristics, enabling targeted sales strategies and customer prioritization.

Enhance Decision-Making Through Business Intelligence:

Design interactive Power BI dashboards that present key performance indicators (KPIs), trends, and analytical insights to support timely, data-driven managerial decision-making

Data Description and Understandings

Data Dictionary

Payment No.	Salesmen	Type	Payment Date	Customer Name	Quauntity	Amount	Discount	Location	Value Category
CInv0002001636	محمد احمد علي	فحم فيتنامي 10 كجم - 7 ساعات	7/29/2025	مؤسسة بيت الوافر للتجارة	200	9200	0	الشمالية	Large
CInv0002001637	محمد احمد علي	فحم فيتنامي 10 كجم - 7 ساعات	7/29/2025	شركة خيمة الضيافة للمحامص - فرع الراجحي	200	9200	0	الشمالية	Large
CInv0002001638	محمد احمد علي	فحم فيتنامي 10 كجم - 7 ساعات	7/30/2025	شركة أسواق لؤلؤة الوادين للمواد الغذائية	50	2300	0	الشمالية	Large
CInv0002001639	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (6 رول)	8/21/2025	محامص حضارة الجنوب للمكسرات - تبوك	30	231	0	الشمالية	Meduim
CInv0002001640	محمد احمد علي	فحم بخور (مقاس 22) - كرتون 30 علبه (10 رول)	8/21/2025	مؤسسة منازل الحجاز التجارية - فرع النهضة	120	936	0	الشمالية	Large
CInv0002001640	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (2 رول)	8/21/2025	مؤسسة منازل الحجاز التجارية - فرع النهضة	90	342	0	الشمالية	Meduim
CInv0002001640	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (6 رول)	8/21/2025	مؤسسة منازل الحجاز التجارية - فرع النهضة	120	924	0	الشمالية	Large
CInv0002001640	محمد احمد علي	فحم بخور (مقاس 50) - 6 رول	8/21/2025	مؤسسة منازل الحجاز التجارية - فرع النهضة	72	1281.6	0	الشمالية	Large
CInv0002001640	محمد احمد علي	فحم بخور (مقاس 50) - كرتون 30 علبه (2 رول)	8/21/2025	مؤسسة منازل الحجاز التجارية - فرع النهضة	60	462	0	الشمالية	Meduim
CInv0002001641	محمد احمد علي	فحم بخور (مقاس 22) - كرتون 30 علبه (10 رول)	8/21/2025	شركة بصمة التوفير للتجارة	60	430.2	0	الشمالية	Meduim
CInv0002001641	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (2 رول)	8/21/2025	شركة بصمة التوفير للتجارة	60	204	0	الشمالية	Meduim
CInv0002001641	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (6 رول)	8/21/2025	شركة بصمة التوفير للتجارة	90	645.3	0	الشمالية	Large
CInv0002001641	محمد احمد علي	فحم بخور (مقاس 50) - 6 رول	8/21/2025	شركة بصمة التوفير للتجارة	54	914.76	0	الشمالية	Large
CInv0002001641	محمد احمد علي	فحم بخور (مقاس 50) - كرتون 30 علبه (2 رول)	8/21/2025	شركة بصمة التوفير للتجارة	60	430.2	0	الشمالية	Meduim
CInv0002001642	محمد احمد علي	فحم بخور (مقاس 22) - كرتون 30 علبه (10 رول)	8/23/2025	مؤسسة درة المحارب للمواد الغذائية - تبوك	30	234	0	الشمالية	Meduim
CInv0002001642	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (6 رول)	8/23/2025	مؤسسة درة المحارب للمواد الغذائية - تبوك	30	231	0	الشمالية	Meduim
CInv0002001642	محمد احمد علي	فحم بخور (مقاس 50) - 6 رول	8/23/2025	مؤسسة درة المحارب للمواد الغذائية - تبوك	18	320.4	0	الشمالية	Meduim
CInv0002001642	محمد احمد علي	فحم بخور (مقاس 50) - كرتون 30 علبه (2 رول)	8/23/2025	مؤسسة درة المحارب للمواد الغذائية - تبوك	30	231	0	الشمالية	Meduim
CInv0002001643	محمد احمد علي	فحم بخور (مقاس 33) - كرتون 30 علبه (6 رول)	8/23/2025	معمل روائح النكهة التجارية - تبوك	30	231	0	الشمالية	Meduim

6852 Row & 10 Columns

Column Name	Description	Type	Use
Payment_Number	Unique identifier for each sales invoice	Categorical	Reference key for transaction tracking; not used directly in modeling
Payment_Date	Date when the sales transaction occurred	Date	Time-series analysis, trend identification, and seasonal patterns
Customer_Name	Name of the customer	Categorical	Customer-level aggregation and segmentation
Type	Product sold in the transaction	Categorical	Product performance analysis and sales distribution
Quantity	Number of units sold	Numerical	Key predictor variable for sales amount
Discount	Discount applied to the transaction	Numerical	Used to analyze the impact of discounts on sales performance
Sales_Amount	Total value of the sales transaction	Numerical	Target variable for predictive modeling
Salesmen	Salesperson responsible for the transaction	Categorical	Sales performance comparison and KPI analysis

Location	Geographic location of the customer or transaction	Categorical	Regional sales analysis and geographic performance comparison
Value_Category	Categorization of sales value (e.g., High, Medium, Low)	Categorical	Sales segmentation, classification analysis, and dashboard filtering

Payment No.	Payment Date	Customer Code	Customer Name	Sales Amount	Payment Status	Area	Salesmen
CslR0010000304	7/6/2025	WH-W-01903	عميل عجز JED Van 4	240.67	Cash	ELQ Van 7	Store Keeper
CslR0010000305	7/16/2025	WH-W-01903	عميل عجز JED Van 4	65.18	Cash	ELQ Van 7	Store Keeper
CslR0010000310	10/23/2025	WH-W-01903	عميل عجز JED Van 4	312.54	Cash	ELQ Van 7	Store Keeper
CslR0012000137	7/30/2025	WH-E-01461	عميل عجز DMM Van 3	16.3	Cash	DMM Van 3	Store Keeper
CslR0012000140	9/2/2025	WH-E-01461	عميل عجز DMM Van 3	43.52	Cash	DMM Van 3	Store Keeper
CslR0014000691	7/9/2025	WH-E-01460	عميل عجز DMM Van 2	23.29	Cash	DMM Van 2	Store Keeper
CslR0022000527	7/31/2025	WH-S-01018	عميل عجز KMX Van 3	529.19	Cash	KMX Van 3	Store Keeper
CslR0026000258	7/1/2025	WH-C-05952	عميل عجز RUH Van 10	11.97	Cash	RUH Van 10	Store Keeper
CslR0026000272	7/14/2025	WH-C-05952	عميل عجز RUH Van 10	9.5	Cash	RUH Van 10	Store Keeper
CslR0037000270	7/31/2025	WH-S-01020	عميل عجز KMX Van 5	115.18	Cash	KMX Van 5	Store Keeper
CslR0037000271	8/21/2025	WH-S-01020	عميل عجز KMX Van 5	186.37	Cash	KMX Van 5	Store Keeper
CslR0044000087	7/31/2025	WH-S-01476	عميل عجز KMX Van 7	21.62	Cash	KMX Van 7	Store Keeper
CslR0044000089	8/18/2025	WH-S-01476	عميل عجز KMX Van 7	101.91	Cash	KMX Van 7	Store Keeper
CslR0051000168	7/6/2025	WH-C-06617	عميل عجز ELQ Van 8	1146.32	Cash	ELQ Van 8	Store Keeper
CslR0051000171	8/2/2025	WH-C-06617	عميل عجز ELQ Van 8	144.9	Cash	ELQ Van 8	Store Keeper
REC0044001079	7/12/2025	WH-S-00055	النخلتين للمعسلات والشيش - شركة توب كيف للتجارة	5313	Cash	KMX Van 7	احمد العتلي
REC0044001212	8/22/2025	WH-S-00055	النخلتين للمعسلات والشيش - شركة توب كيف للتجارة	3684	Cash	KMX Van 7	احمد العتلي
REC0044001239	8/26/2025	WH-S-00055	النخلتين للمعسلات والشيش - شركة توب كيف للتجارة	0.6	Cash	KMX Van 7	احمد العتلي
REC0044001346	9/27/2025	WH-S-00055	النخلتين للمعسلات والشيش - شركة توب كيف للتجارة	3275.2	Cash	KMX Van 7	احمد العتلي

4054 Row & 8 Columns

Column Name	Description	Type	Use
Payment_Number	Unique identifier for each sales invoice	Categorical	Reference key for transaction tracking; not used directly in modeling
Payment_Date	Date when the sales transaction occurred	Date	Time-series analysis, trend identification, and seasonal patterns
Customer_Name	Name of the customer	Categorical	Customer-level aggregation and segmentation
Sales_Amount	Total value of the sales transaction	Numerical	Target variable for predictive modeling
Salesmen	Salesperson responsible for the transaction	Categorical	Sales performance comparison and KPI analysis
Customer_Code	Unique code assigned to each customer	Categorical	Primary key for sales
Area (Van Type)	Type of delivery van used for the transaction	Categorical	Operational performance analysis
Payment_Status (Payment Method)	Method of payment used by the customer (Cash, ATM, Wire Transfer)	Categorical	Analysis of payment preferences and their impact on sales and collections
Category Level	Categorization of sales value (e.g., High, Medium, Low)	Categorical	Sales segmentation, classification analysis, and dashboard filtering

Exploratory Data Analysis (EDA)

Initial Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, and behavior of the sales and collections data before applying predictive and clustering models. Various charts and statistical summaries were used to uncover patterns, trends, and anomalies relevant to the project goals.

Histograms and descriptive statistics were applied to numerical variables such as Sales Amount, Quantity, and Discount. The analysis showed that sales amount follows a right-skewed distribution, indicating that most transactions are low to medium in value, while a small number of transactions contribute to very high sales. Box plots for quantity and discount revealed the presence of extreme outliers, particularly in discount values, suggesting occasional large discounts applied to specific transactions.

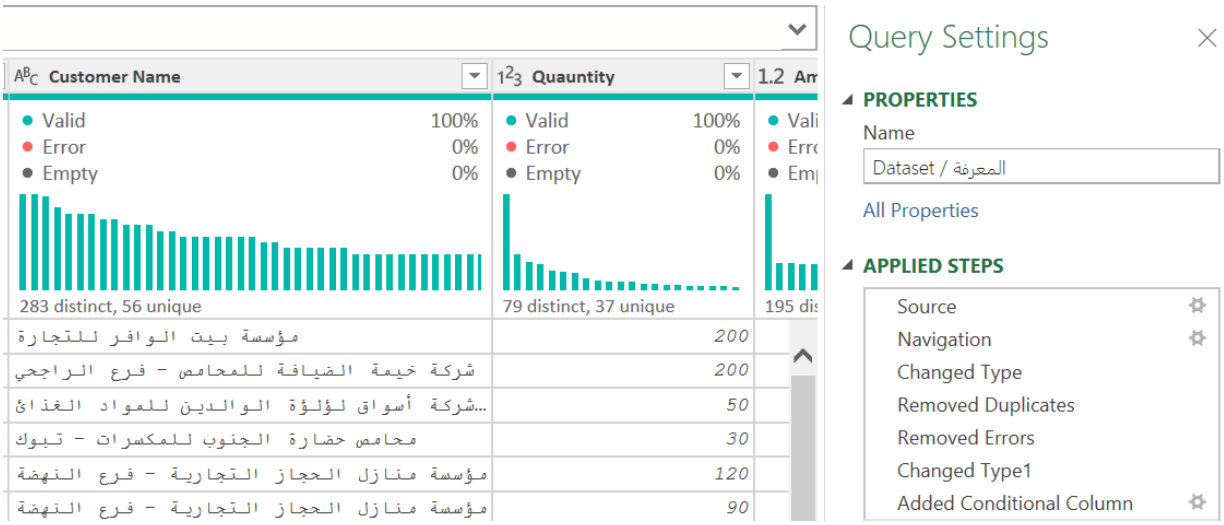
Bar charts and pivot tables were used to analyze categorical variables such as Location, Value Category, and Payment Method. The results showed that certain regions contribute more significantly to total sales, while large value category transactions, although fewer, account for a substantial portion of revenue. Payment method analysis highlighted differences in customer payment behavior, which is relevant for evaluating collection efficiency.

Scatter plots between Quantity and Sales Amount indicated a strong positive relationship, confirming that higher quantities generally lead to higher sales values. However, variations caused by discounting behavior justified the need for predictive modeling to accurately estimate sales amounts.

Overall, the EDA phase provided valuable insights into key business drivers, identified data variability and outliers, and guided feature selection for prediction and clustering models. These findings directly support the project's objective of improving sales forecasting, customer segmentation, and data-driven decision-making through business intelligence.

Data Primary Cleaning and Transformation

is the process of identifying and correcting errors, inconsistencies, or missing values in a dataset to improve its quality and reliability. It includes removing duplicates, fixing incorrect entries, handling missing data, and formatting values properly. This step ensures that the data is accurate, consistent, and ready for meaningful analysis and modeling.



1. **Changed Data Types**
   Data types (numeric, text, date) were reviewed and correctly assigned to each column to ensure consistency and proper analysis.

2. **Removed Irrelevant Columns**
   Unnecessary columns such as IDs, row numbers, and non-analytical fields were removed to keep only features relevant to analysis and modeling.

3. **Removed Errors**
   Rows containing invalid values or data errors were identified and removed to maintain data accuracy.

4. **Removed Duplicate Records**
   Duplicate rows were eliminated to prevent biased analysis and ensure reliable model results.

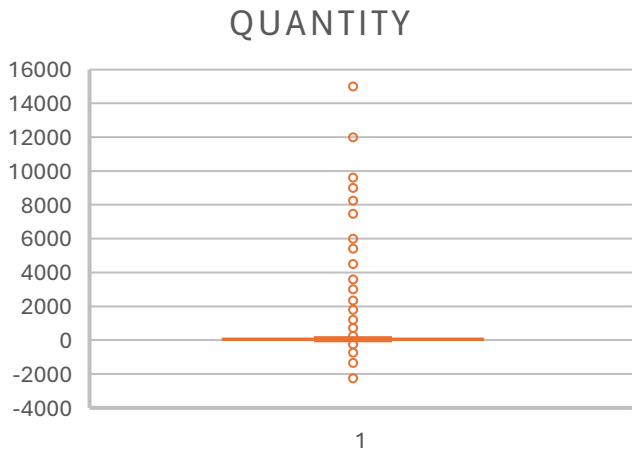
5. **Removed Blank Rows**
   Empty or incomplete rows were removed to maintain a clean and consistent dataset structure.

Purpose:
These steps ensured that the dataset was clean, accurate, and well-structured, making it ready for descriptive analysis, predictive modeling, and clustering.

Data Visualization and Insights

Columns Distributions

Quauntity	
Mean	127.4245
Standard Error	6.048213
Median	30
Mode	30
Standard Deviation	500.6518
Sample Variance	250652.2
Kurtosis	274.1871
Skewness	14.00002
Range	17250
Minimum	-2250
Maximum	15000
Sum	873113
Count	6852



Quantity Distribution Analysis

The Quantity chart illustrates the distribution of sold quantities across all transactions. Visualization shows that most transactions involve small quantities, while a limited number of transactions have very large quantities, creating several visible outliers.

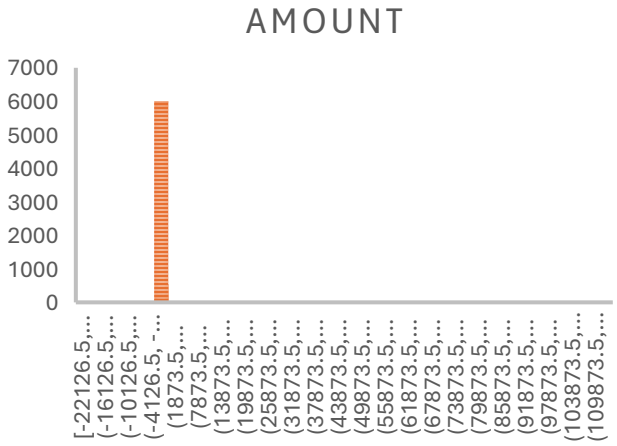
From the descriptive statistics, the mean quantity (127.42) is much higher than the median and mode (both 30), indicating that the data is highly right-skewed. This is confirmed by the skewness value of 14.00, which shows that extreme high-quantity sales significantly influence the average. Additionally, the very high kurtosis (274.19) suggests the presence of extreme outliers and a heavy-tailed distribution.

The standard deviation (500.65) and wide range (17,250) further highlight the high variability in sales quantities. The minimum value of –2,250 may represent returns or data corrections, while the maximum value of 15,000 reflects bulk or wholesale transactions.

Key Insight

Most sales occur in relatively small quantities, but a small number of bulk transactions heavily impact on overall sales volume. This pattern is important for understanding sales behavior, identifying bulk buyers, and designing predictive models, as these extreme values can significantly affect averages and model performance

Amount	
Mean	1209.634438
Standard Error	53.37989474
Median	320.4
Mode	231
Standard Deviation	4418.617317
Sample Variance	19524178.99
Kurtosis	187.3597769
Skewness	11.68258623
Range	137184.3
Minimum	-22126.5
Maximum	115057.8
Sum	8288415.17
Count	6852



Sales Amount Distribution Analysis

The Sales Amount chart illustrates the distribution of sales values across all transactions. The visualization shows that most transactions occur at relatively low sales amounts, while a small number of transactions have extremely high values, resulting in several prominent outliers.

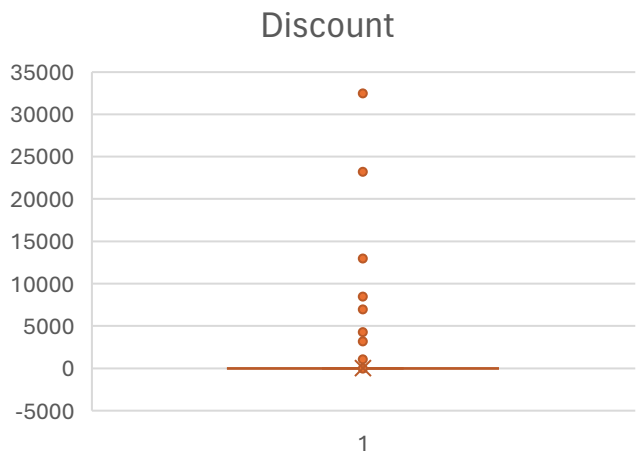
From the descriptive statistics, the mean sales amount (1,209.63) is significantly higher than the median (320.4) and mode (231), indicating a strongly right-skewed distribution. This is supported by the high skewness value of 11.68, which confirms that large sales transactions disproportionately increase the average sales amount. Additionally, the very high kurtosis (187.36) suggests a heavy-tailed distribution with extreme outliers.

The large standard deviation (4,418.62) and wide range (137,184.3) highlight substantial variability in sales performance. The minimum value of –22,126.5 likely reflects returns, refunds, or data adjustments, while the maximum value of 115,057.8 represents high-value or bulk sales transactions.

Key Insight

Most sales transactions generate relatively low revenue, but a small number of high-value sales contribute significantly to total revenue. This distribution is critical for understanding revenue concentration, identifying key high-value customers, and building predictive models, as extreme values can strongly influence overall performance metrics and model accuracy.

Discount	
Mean	**25.44118506**
Standard Error	6.59203832
Median	**0**
Mode	**0**
Standard Deviation	**545.6678926**
Sample Variance	**297753.449**
Kurtosis	**2365.564643**
Skewness	**45.47155096**
Range	32470
Minimum	-18
Maximum	32452
Sum	174323
Count	6852



Discount Distribution Analysis

The Discount chart shows the distribution of discount values applied across all transactions. The visualization indicates that most transactions have no discount applied, while a small number of transactions include very large discounts, creating extreme outliers.

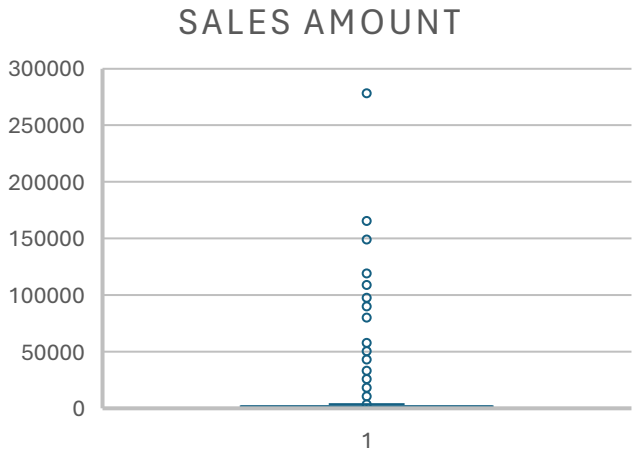
From the descriptive statistics, the mean discount (25.44) is much higher than the median and mode (both 0), which clearly indicates a highly right-skewed distribution. This is strongly supported by the skewness value of 45.47, showing that a few exceptionally large discounts heavily influence the average. The very high kurtosis (2365.56) further confirms the presence of extreme outliers and a highly peaked distribution with heavy tails.

The standard deviation (545.67) and the wide range (32,470) highlight substantial variability in discount values. The minimum value of –18 may represent discount reversals or data adjustments, while the maximum value of 32,452 suggests exceptional promotional or bulk discount cases.

Key Insight

Most sales transactions are completed without discounts, but a very small number of heavily discounted transactions significantly affect the overall discount statistics. This pattern is important for sales and pricing analysis, as extreme discount values can influence revenue, profitability, and predictive model performance.

Sales Amount	
Mean	**2906.07668**
Standard Error	169.0108956
Median	**634.11**
Mode	**265.65**
Standard Deviation	**10761.09772**
Sample Variance	**115801224.2**
Kurtosis	**179.4096393**
Skewness	**11.03098461**
Range	278374.99
Minimum	0.01
Maximum	278375
Sum	11781234.86
Count	4054



Sales Amount Distribution Analysis

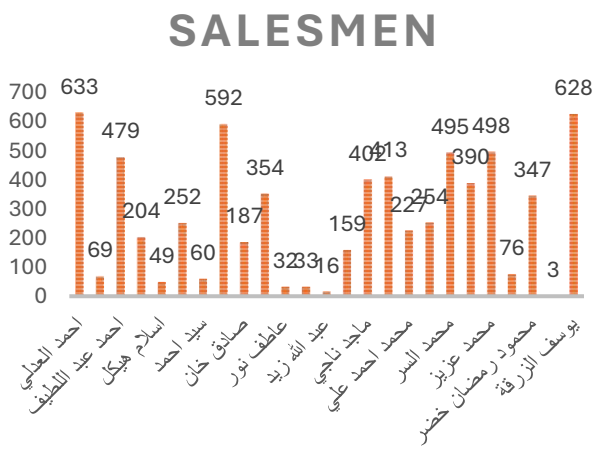
The Sales Amount chart shows the distribution of sales values across all transactions. The visualization indicates that most sales transactions are relatively low in value, while a small number of transactions have extremely high sales amounts, resulting in several clear outliers.

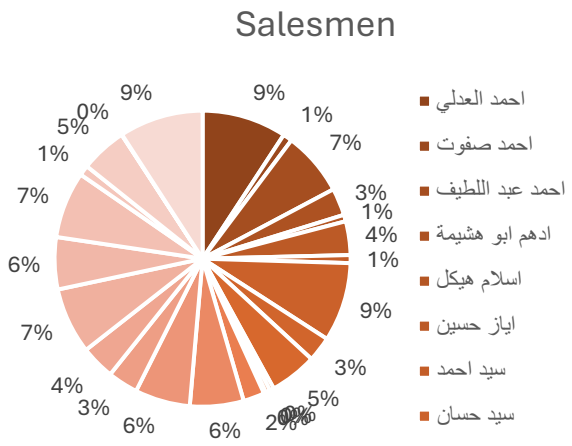
From the descriptive statistics, the mean sales amount (2,906.08) is significantly higher than the median (634.11) and mode (265.65), which indicates a highly right-skewed distribution. This is confirmed by the skewness value of 11.03, showing that a few very large sales transactions heavily influence the average sales value. The very high kurtosis (179.41) further suggests the presence of extreme outliers and a heavy-tailed distribution.

The large standard deviation (10,761.10) and wide range (278,374.99) reflect substantial variability in sales amounts. The minimum value (0.01) represents very small transactions, while the maximum value (278,375) is likely to correspond to major bulk or high-value sales.

Key Insight

Most sales transactions generate relatively low revenue, but a small number of high-value sales contribute disproportionately to total revenue. This pattern is critical for sales analysis and predictive modeling, as these extreme values can strongly impact averages, forecasting accuracy, and overall business performance.





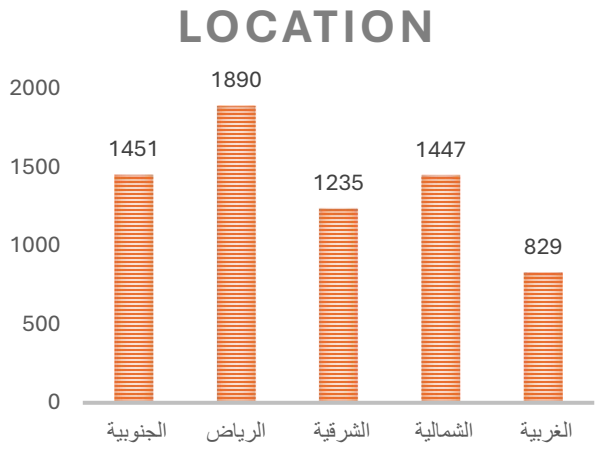
Salesmen Distribution Analysis

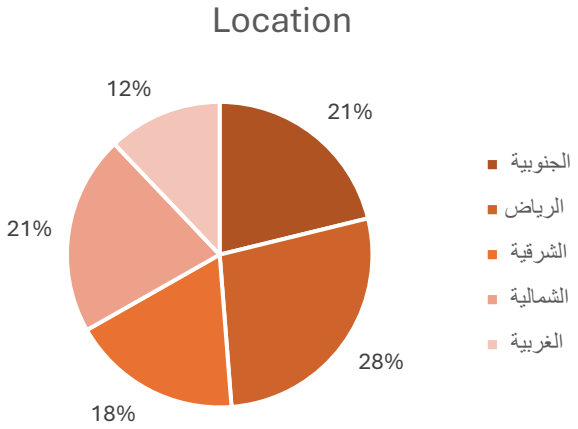
The Salesmen charts illustrate the distribution of transactions handled by each salesperson, measured by the number of recorded sales rather than sales value. The bar chart shows noticeable variation in transaction counts across salesmen, with a small group handling a significantly higher number of transactions, while others contribute relatively fewer sales. This indicates an imbalance in workload or customer allocation among the sales team.

The pie chart further emphasizes this distribution by showing each salesperson's percentage contribution to the total number of transactions. A few salesmen account for the largest shares (around 7–9% each), while several others represent very small proportions (1–3%). This long-tail pattern suggests that sales activity is concentrated among a limited number of high-performing or highly active sales representatives.

Key Insight

Sales performance, when measured by transaction count, is unevenly distributed among salesmen. A small group dominates sales activity, which may reflect differences in territory coverage, customer portfolios, or experience levels. This insight is important for workforce planning, performance evaluation, and designing fair incentive or support strategies, as it highlights potential over-reliance on top performers and underutilization of others.





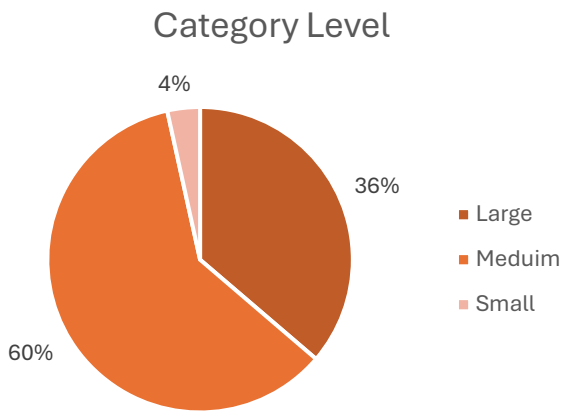
Location Distribution Analysis

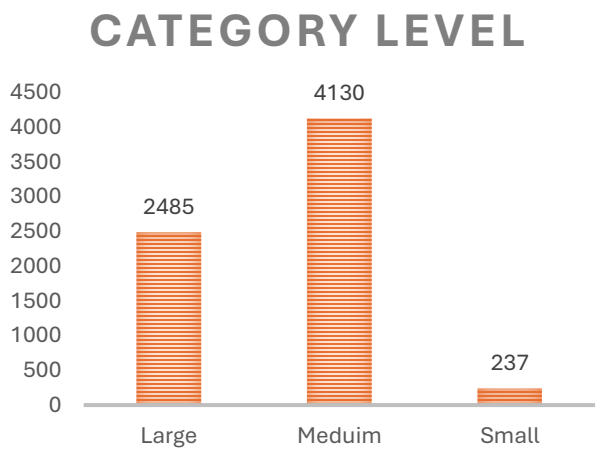
The Location charts present the distribution of transactions across different geographic regions based on transaction count. The bar chart shows that Riyadh has the highest number of transactions (1,890), followed by the Southern region (1,451) and the Northern region (1,447). The Eastern region accounts for 1,235 transactions, while the Western region has the lowest count at 829.

The pie chart further highlights the proportional contribution of each location. Riyadh represents the largest share of transactions at approximately 28%, followed by the Southern and Northern regions, each contributing about 21%. The Eastern region accounts for roughly 18%, while the Western region contributes the smallest share at 12%.

Key Insight

Sales activity is geographically concentrated, with Riyadh emerging as the dominant market in terms of transaction volume. However, the Southern and Northern regions also show strong activity, indicating diversified regional demand. These insights are valuable for regional sales planning, resource allocation, and targeted marketing strategies, as different regions contribute unevenly to overall business performance.





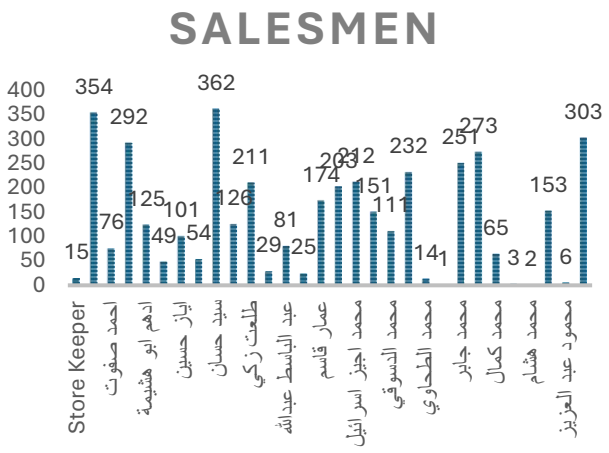
Category Level Distribution Analysis

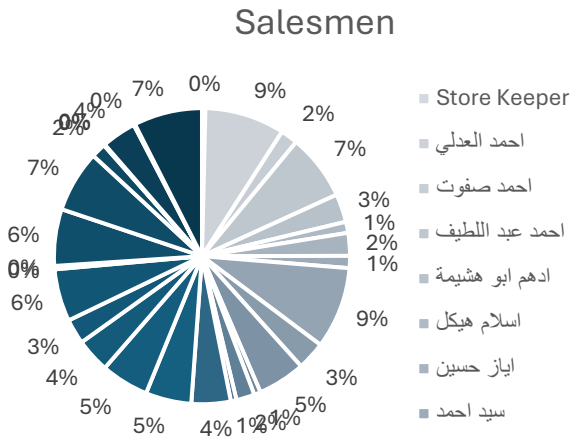
The Category Level charts present the distribution of transactions across different value categories (Large, Medium, and Small) based on transaction count. The bar chart shows that the *Medium* category dominates the dataset with the highest number of transactions, followed by the *Large* category, while the *Small* category represents only a very small portion of total sales activity. This indicates that most transactions fall within a mid-range value rather than extremely high or low values.

The pie chart reinforces this pattern by illustrating the proportional contribution of each category. Approximately 60% of transactions belong to the *Medium* category, around 36% are *Large*, and only about 4% are *Small*. This skewed distribution highlights a clear concentration of sales activity in medium-value transactions, with low-value sales occurring infrequently.

Key Insight

Sales activity is heavily concentrated in the Medium value category, suggesting that the company's core business relies on mid-sized transactions. Large transactions still play a significant role, while small-value sales contribute minimally. This insight is important for pricing strategy, customer segmentation, and sales forecasting, as it helps focus operational and analytical efforts on the categories that drive the majority of transaction volume





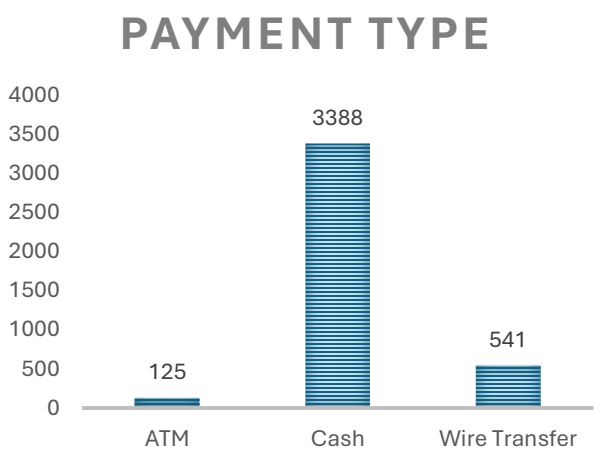
Salesmen Distribution Analysis

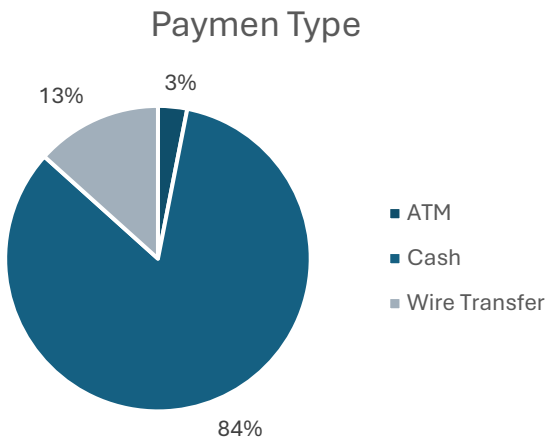
The Salesmen charts present the distribution of sales transactions based on **transaction count** for each salesperson. The bar chart shows clear variation in the number of transactions handled by different salesmen. A few salesmen record relatively high transaction counts, while many others handle a moderate or low number of sales, indicating uneven sales activity across the team.

The pie chart reinforces this pattern by illustrating each salesperson's percentage contribution to the total number of transactions. Several salesmen contribute larger shares (around 7–9%), whereas others account for very small portions (1–3%), creating a long-tail distribution.

Key Insight

Sales activity, measured by transaction count, is concentrated among a limited number of salesmen. This suggests differences in customer allocation, territory coverage, or experience levels. Understanding this imbalance is important for evaluating salesperson workload, improving sales resource distribution, and designing fair performance management and incentive strategies.





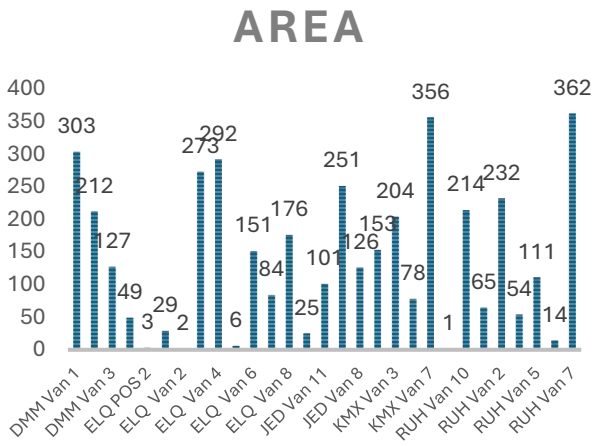
Payment Type Distribution Analysis

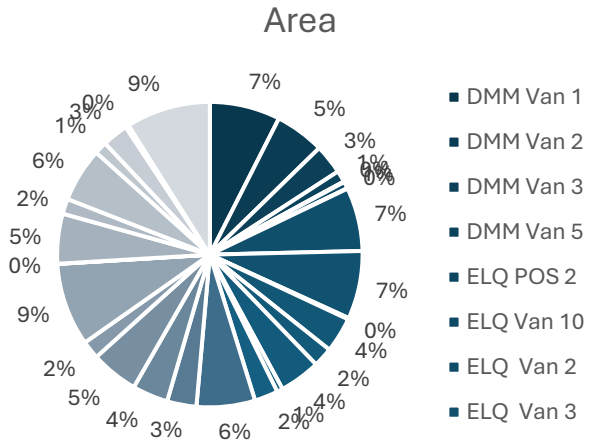
The Payment Type charts show how transactions are distributed across different payment methods based on the number of transactions. The bar chart indicates that Cash is by far the most frequently used payment method, with 3,388 transactions, compared to Wire Transfer (541) and ATM (125) transactions. This highlights a strong customer preference for cash payments.

The pie chart further reinforces this pattern, showing that Cash accounts for approximately 84% of all transactions, while Wire Transfer represents about 13% and ATM only around 3%. The dominance of cash payments suggests limited adoption of electronic or bank-based payment methods among customers.

Key Insight

The organization relies heavily on cash transactions, which may increase operational risks and handling costs. This insight is important for financial planning and process improvement, as it highlights opportunities to encourage more secure and efficient payment methods such as wire transfers or electronic payments.





Area Distribution Analysis

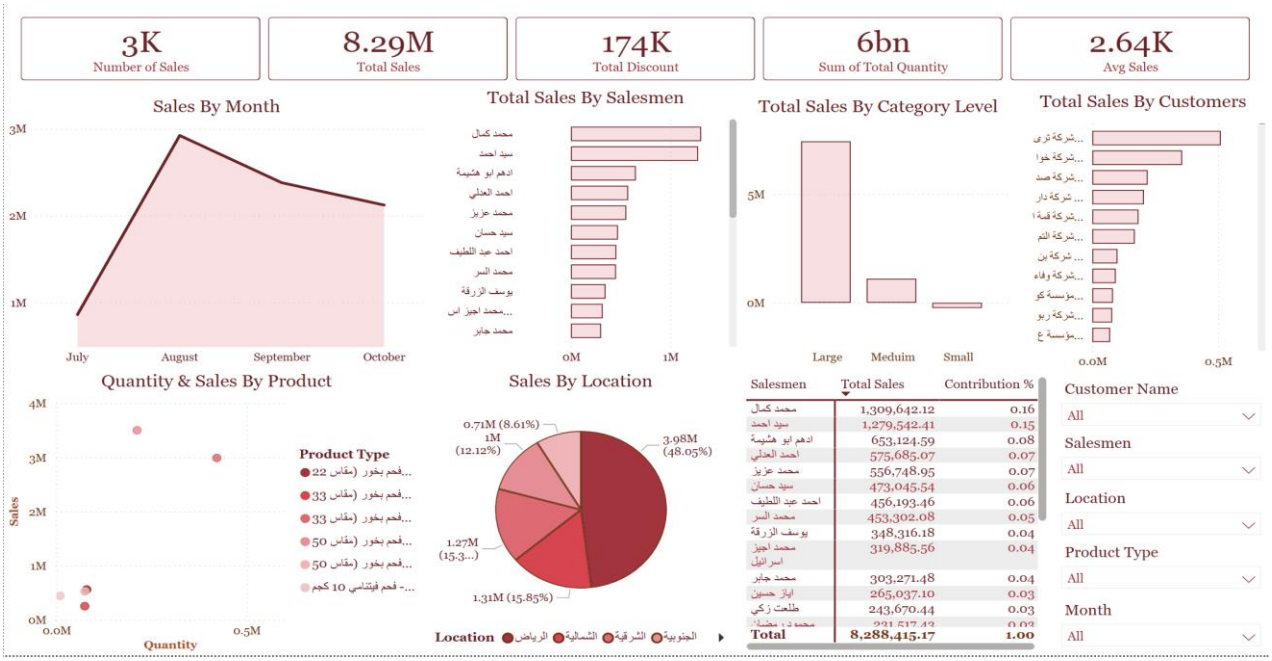
The Area charts present the distribution of sales transactions across different operational areas (vans/points), measured by transaction count rather than sales value. The bar chart shows clear variation among areas, where a few areas (such as DMM and RUH vans) record noticeably higher numbers of transactions, while many other areas contribute relatively smaller counts. This indicates that sales activity is not evenly distributed across all operational units.

The pie chart reinforces this observation by highlighting the percentage share of each area in total transactions. A limited number of areas account for the largest proportions (around 7–9% each), whereas several areas represent very small shares (0–3%). This long-tail distribution suggests concentration of sales activity in specific high-demand or high-coverage areas.

Key Insight

Sales transactions are concentrated in a small number of areas, while many areas are underutilized. This insight is valuable for operational planning, as it can support decisions related to route optimization, resource allocation, and balancing workload across areas to improve overall efficiency and coverage.

Power BI Dashboards



Purpose

This dashboard provides a comprehensive overview of sales performance by time, salesperson, customer, product category, location, and quantity, enabling management to monitor trends, identify top contributors, and support data-driven decisions.

Business Question 1: How is overall sales performance evolving over time?

Chart: *Sales by Month (Area Chart)*

Description:

This chart displays total sales amounts aggregated by month (July to October).

Insight Derived:

Sales increased sharply from July to August, peaked in August, and then gradually declined through September and October. This suggests a seasonal or campaign-driven surge in August, followed by a slowdown. The pattern highlights the importance of understanding monthly demand fluctuations for inventory planning and sales forecasting.

Business Question 2: Who are the top-performing salesmen?

Chart: *Total Sales by Salesmen (Horizontal Bar Chart)*

Description:
This chart ranks salesmen based on their total sales contribution.

Insight Derived:
A small group of salesmen contributes a disproportionately large share of total sales, while the majority have relatively lower contributions. This indicates performance concentration among top sellers and suggests opportunities for training, redistribution of customer accounts, or incentive optimization for lower-performing salesmen.

Business Question 3: Which product value categories generate the most sales?

Chart: *Total Sales by Category Level*

Description:
This bar chart compares total sales across product value categories (Large, Medium, Small).

Insight Derived:
Large-category products dominate total sales, while Medium contributes moderately and Small has minimal impact. This suggests that high-value products are the primary revenue drivers, emphasizing the need to prioritize their availability, promotion, and supply chain efficiency.

Business Question 4: Which customers generate the highest sales?

Chart: *Total Sales by Customers*

Description:
This chart shows the top customers ranked by total sales amount.

Insight Derived:
Sales are concentrated among a limited number of key customers. This highlights the importance of customer retention strategies for high-value clients and the potential risk of revenue dependency on a small customer base.

Business Question 5: How does quantity relate to sales value across products?

Chart: *Quantity vs Sales by Product (Scatter Plot)*

Description:
This scatter plot visualizes the relationship between quantity sold and sales amount for different product types.

Insight Derived:
Higher quantities generally correspond to higher sales values, but the relationship is not perfectly linear. Some products generate high sales with relatively low quantities, indicating premium pricing, while others require bulk sales to generate revenue. This insight supports differentiated pricing and sales strategies by product type.

Business Question 6: How are sales distributed across locations?

Chart: *Sales by Location (Pie Chart)*

Description:
This pie chart displays the percentage contribution of each location to total sales.

Insight Derived:
One location contributes nearly half of total sales, while the remaining locations share the rest. This geographical concentration suggests strong regional demand in specific areas and highlights where expansion or marketing efforts may yield the highest returns.

Business Question 7: What are the key performance indicators at a glance?

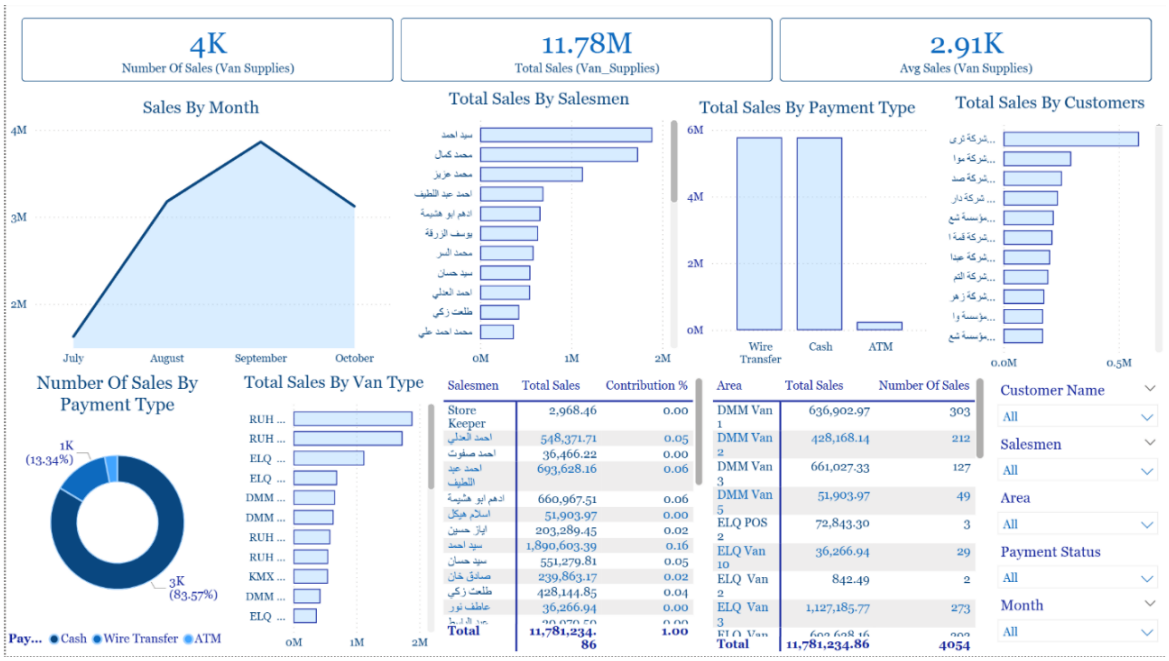
KPIs (Top Cards)

- **Number of Sales**, **Total Sales**, **Total Discount**, **Total Quantity** & **Average Sales**

Insight Derived:
These KPIs provide an immediate summary of business performance. The high total quantity alongside moderate average sales suggests frequent small-to-medium transactions, while total discounts indicate pricing and promotion intensity affecting profitability.

Overall Dashboard Insight

The dashboard reveals that sales performance is highly concentrated across specific months, products, locations, customers, and salesmen. High-value products and top-performing sales representatives drive most of the revenue, while significant variability exists across regions and customer segments. These insights support strategic decisions related to sales planning, customer prioritization, pricing strategies, and operational optimization.



Purpose

This dashboard presents a focused analysis of **Van Supplies sales performance**, providing insights by time, salesperson, payment type, customer, van type, and area. It supports management in tracking trends, identifying key contributors, and optimizing operational and sales strategies related to van-based distribution.

Business Question 1: How are Van Supplies sales changing over time?

Chart: Sales by Month (Area Chart)
Description:
This chart shows total Van Supplies sales aggregated monthly from July to October.

Insight Derived:

Sales increased significantly from July to August, reached their highest level in September, and then declined in October. This trend suggests strong demand growth toward the late summer period followed by a slowdown, possibly due to seasonal demand or reduced distribution activity. Understanding this pattern is essential for planning van routes, inventory loading, and monthly sales targets.

Business Question 2: Which salesmen generate the highest Van Supplies sales?

Chart: Total Sales by Salesmen (Horizontal Bar Chart)
Description:
This chart ranks salesmen based on their total Van Supplies sales contribution.

Insight Derived:
A limited number of salesmen account for a large portion of Van Supplies revenue, while others contribute smaller shares. This highlights performance differences among sales representatives and suggests opportunities for knowledge sharing, targeted training, or workload rebalancing to improve overall sales efficiency.

Business Question 3: How are Van Supplies sales distributed by payment type?

Chart: Total Sales by Payment Type (Bar Chart)
Description:
This chart compares Van Supplies sales across different payment methods (Cash, Wire Transfer, ATM).

Insight Derived:
Cash and wire transfer dominate Van Supplies transactions, while ATM usage is minimal. This indicates a strong preference for immediate or direct payment methods in van-based sales, which may simplify cash flow management but also presents opportunities to promote digital payment adoption.

Business Question 4: Which customers contribute the most to Van Supplies sales?

Chart: Total Sales by Customers
Description:
This chart ranks customers based on their total Van Supplies purchases.

Insight Derived:
Van Supplies sales are concentrated among a small group of key customers. Maintaining strong relationships with these customers is critical, as losing them could significantly impact revenue. At the same time, there is potential to increase sales by expanding engagement with lower-volume customers.

Business Question 5: How are sales and transaction volume distributed by van type and area?

Charts: Total Sales by Van Type & Area Table (Total Sales & Number of Sales)

Description:

These visuals compare sales performance across different van types and operational areas.

Insight Derived:

Certain van types and areas generate significantly higher sales and transaction volumes, indicating more efficient routes or stronger customer density. This insight supports route optimization, fleet utilization decisions, and targeted expansion in high-performing areas.

Business Question 6: What is the distribution of Van Supplies transactions by payment method?

Chart: Number of Sales by Payment Type (Donut Chart)

Description:

This chart shows the proportion of Van Supplies transactions by payment type based on transaction count.

Insight Derived:

Most transactions are completed using cash, confirming that van-based sales rely heavily on immediate payments. This reinforces the importance of secure cash handling processes and highlights opportunities to gradually introduce alternative payment options.

Business Question 7: What are the key Van Supplies performance indicators at a glance?

KPIs (Top Cards):

- Number of Sales, Total Sales & Average Sales

Insight Derived:

The KPIs provide a quick overview of Van Supplies performance. A high number of sales combined with a moderate average sale value suggests frequent transactions of relatively small to medium size, which aligns with the operational nature of van-based distribution.

Overall Dashboard Insight

The Van Supplies dashboard reveals that sales performance is driven by specific months, a small group of high-performing salesmen, key customers, and certain van routes and areas. Cash-based transactions dominate, and sales activity peaks during late summer. These insights support better planning for van allocation, salesperson performance management, payment strategy optimization, and customer prioritization—ultimately improving operational efficiency and revenue sustainability.

Advanced Analytics and AI Modeling

Supervised Models

To predict sales performance, supervised machine learning models were implemented using RapidMiner:

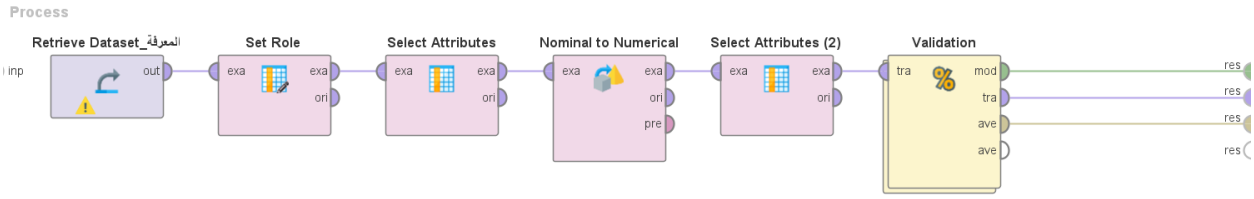
- Linear Regression: Selected as a baseline model due to its simplicity and interpretability. It provides clear insights into the relationship between sales amount and key explanatory variables such as quantity, discount, customer behavior, and time-based factors.
- Random Forest: Chosen for its robustness and strong predictive performance. This ensemble model effectively captures non-linear relationships and complex interactions between features while reducing overfitting, making it suitable for modeling real-world sales data.

Unsupervised Model

Customer segmentation was performed using the K-Means clustering algorithm to identify distinct customer groups based on sales and collections behavior. Key variables used for clustering included total sales amount, number of transactions, total collections, and collection efficiency. Prior to clustering, features were normalized to ensure equal contribution.

Prediction Analysis

is a data analysis technique that uses historical data and statistical or machine-learning models to forecast future outcomes.

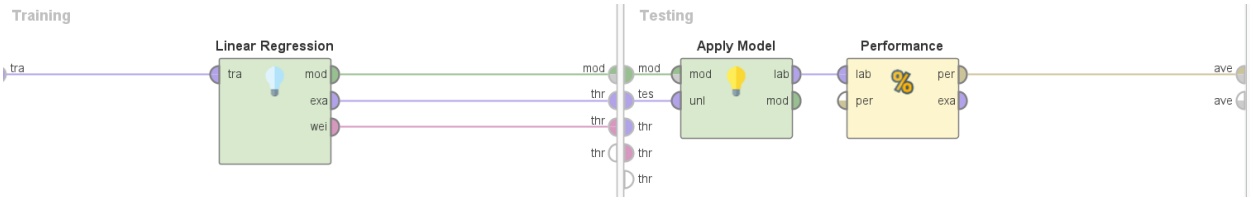


Preparing prediction model:

- Retrieve Dataset: Loads the main dataset
- Set Role: Defines special columns (e.g., ID as *id*, or target of prediction).
- Select Attributes: Keeps only the columns I want to analyze.
- Nominal to Numerical: Converts categorical columns (Location, Value Category, Type) into numbers.
- Select Attributes (2): Final cleanup—handling multi carnality.
- Validation: Tests a model using training/validation splits.

Linear Regression

is a predictive analysis technique used to model the relationship between a dependent variable (Sales Amount) and one or more independent variables (such as Quantity, Discount, Location, and Value Category). It works by fitting a straight line that best represents how changes in the input variables affect sales.



Training phase:

The Linear Regression model learns the relationship between input variables and sales amount from the training data.

Testing phase:

The trained model is applied to unseen data using *Apply Model*, and its performance is evaluated using error metrics in the *Performance* operator.

This process helps assess how accurately the model can predict sales and ensures the reliability of the prediction results.

Correlation (0.893)

The correlation value measures the strength of the linear relationship between the **actual sales amount** and the **predicted sales amount**.
A correlation of **0.893** indicates a **strong positive relationship**, meaning the model's predictions closely follow the actual sales values.

Squared Correlation ($R^2 = 0.798$)

The squared correlation, also known as $\mathbf{R^2}$, shows how much of the variance in sales amount is explained by the model.
An R^2 value of **0.798** means that approximately **79.8% of the variation in sales amount** is explained by the input variables, which reflects a **strong model fit**.

Root Mean Squared Error (RMSE = 2056.34)

RMSE measures the average prediction error in the same unit as the target variable (sales amount).
An RMSE of **2056.34** means that, on average, the model's predictions deviate from the actual sales by about **2056 units**, which is acceptable given the overall sales scale.

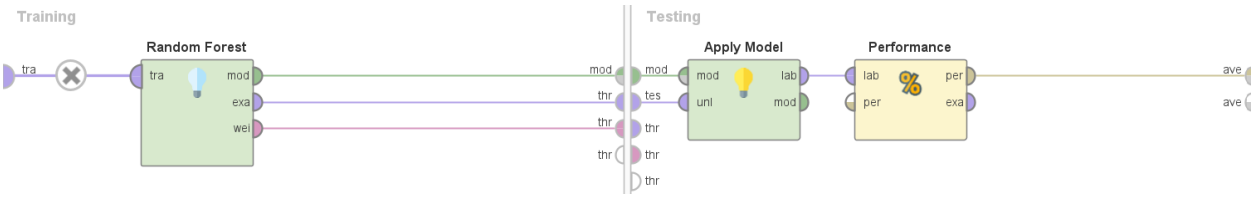
Overall Interpretation

The Linear Regression model demonstrates **strong predictive performance**, with a high correlation and R^2 value, indicating that it effectively captures the relationship between the explanatory variables and sales amount. While some prediction error exists, it remains within an acceptable range for managerial and analytical purposes.

Random Forest

is a supervised machine learning algorithm used for prediction that works by creating multiple decision trees during training and combining their outputs to produce a final prediction. In regression problems, the final prediction is the average of all trees' predictions, which improves accuracy and stability.

In this project, Random Forest is used to predict Sales Amount based on selected numerical and transformed categorical features (such as Quantity, Discount, Location, and Value Category).



Training Phase:

The Random Forest operator is trained using the training dataset. It builds many decision trees using different samples and features to learn complex patterns in the data.

Testing Phase:

The trained Random Forest model is applied to the test dataset to generate predicted sales amounts. Then the performance operator evaluates the model using regression metrics such as correlation and RMSE to measure prediction accuracy.

Correlation (0.948)

The correlation value measures the strength of the relationship between the actual sales amount and the predicted sales amount.
A correlation of **0.948** indicates a **very strong positive relationship**, meaning the Random Forest model's predictions closely track the actual sales values.

Squared Correlation ($R^2 = 0.899$)

The squared correlation (R^2) represents how much of the variation in sales amount is explained by the model.
An **R^2 value of 0.899** means that approximately **89.9% of the variance in sales amount** is explained by the input variables, which indicates an excellent model fit.

Root Mean Squared Error (RMSE = 1372.34)

RMSE measures the average prediction error in the same unit as sales amount.
An RMSE of **1372.34** means that, on average, the model's predictions differ from the actual sales by about **1372 units**, which is lower than the Linear Regression model and indicates improved accuracy.

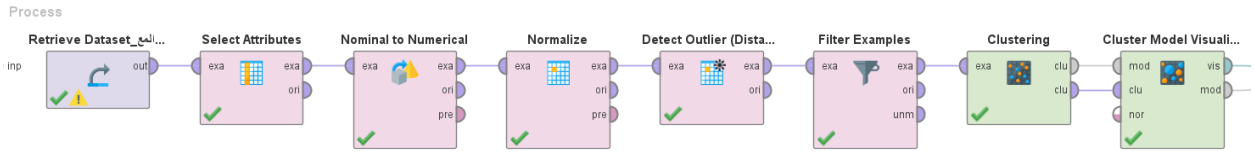
Overall Interpretation

The Random Forest model demonstrates **very strong predictive performance**, outperforming Linear Regression with higher correlation and R^2 values and a lower RMSE. This shows that Random Forest captures complex, non-linear relationships in the data more effectively, making it a highly suitable model for predicting sales amount in this project.

Clustering

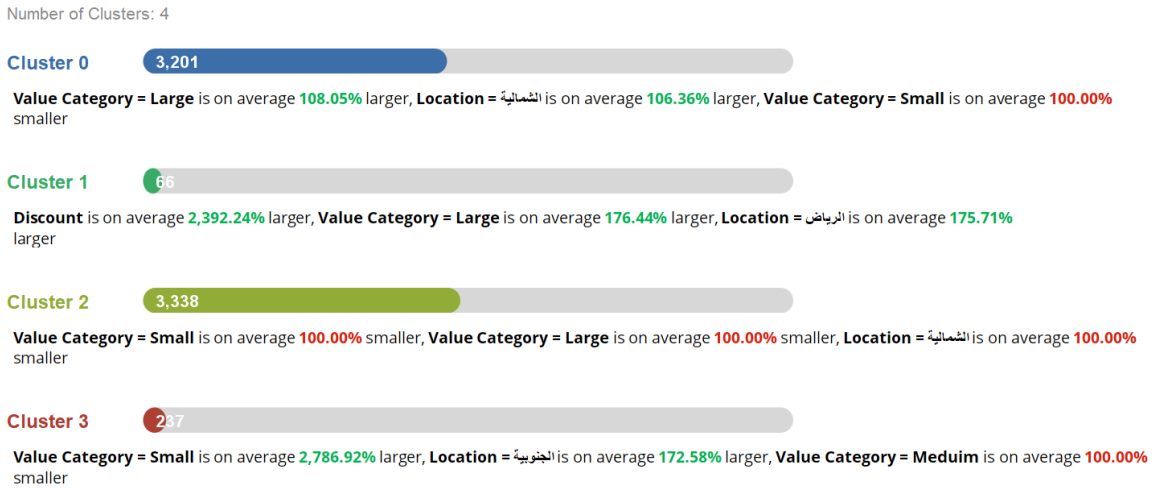
is an unsupervised machine learning technique used to group data points into clusters based on their similarity, without using predefined labels or target variables. The goal of clustering is to ensure that data points within the same cluster are more similar to each other than to those in different clusters.

In a business context, clustering helps identify hidden patterns and natural groupings in data, such as customers with similar purchasing behavior or payment patterns. For example, customers can be clustered based on sales volume, frequency, or collection efficiency to distinguish high-value, medium-risk, and low-priority customer segments.



Clustering Process Explanation

- Retrieve Dataset: The process begins by loading the dataset into RapidMiner..
- Select Attributes: Relevant attributes are selected while irrelevant or non-informative variables are excluded.
- Nominal to Numerical: Categorical (nominal) attributes are converted into numerical format so they can be processed by distance-based clustering algorithms.
- Normalize: All numerical attributes are scaled to a common range.
- Detect Outliers (Distance-Based): Potential outliers are identified based on distance measures.
- Filter Examples: Detected outliers are removed from the dataset.
- Clustering: The cleaned and prepared dataset is clustered using an unsupervised learning algorithm (such as K-Means).
- Cluster Model Visualization: The final step visualizes the clustering results, allowing interpretation of cluster structure, separation, and characteristics.



Clustering Results Interpretation (4 Clusters)

The clustering model grouped the data into four distinct clusters based on similarities in value category, discount behavior, and location.

- **Cluster 0 (3,201 records):**
  Dominated by large-value sales, mainly from the Northern region, with very low presence of small-value transactions. This cluster represents stable, high-value sales behavior.

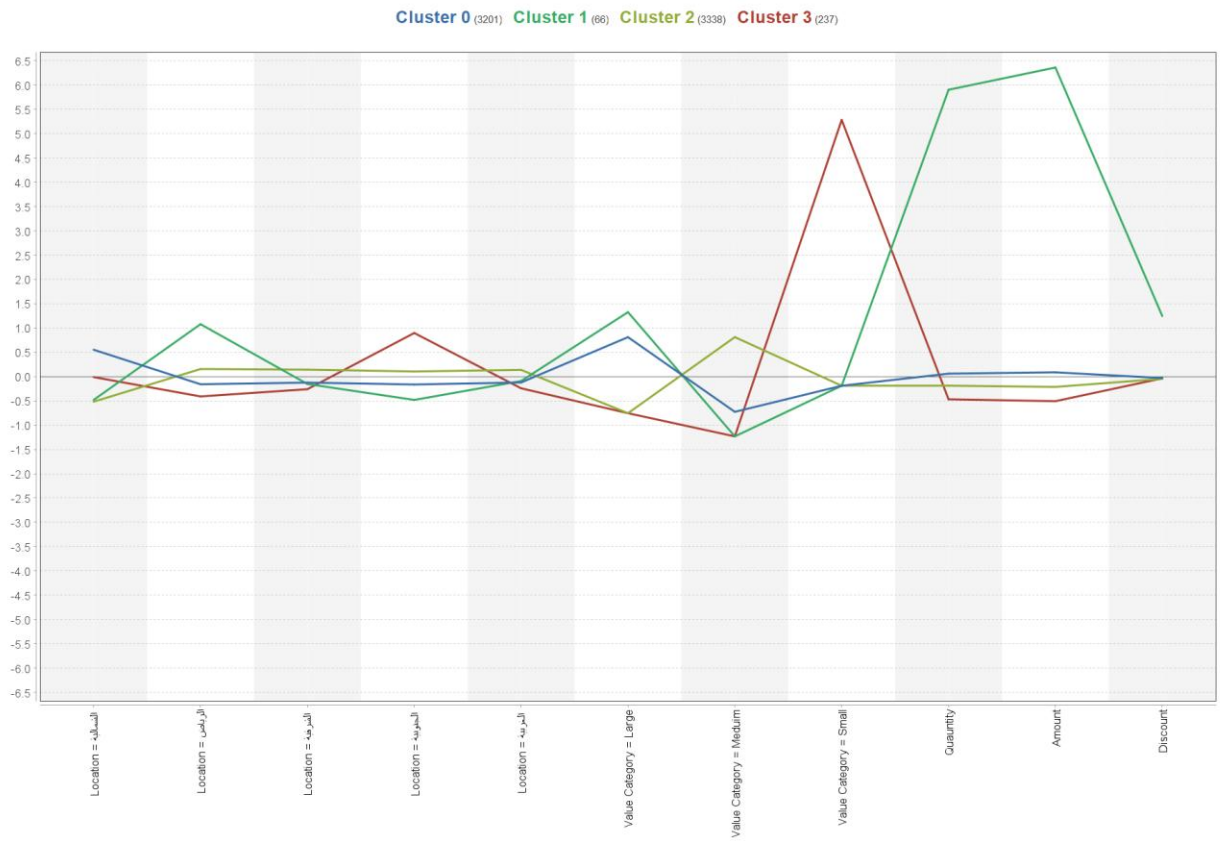
- **Cluster 1 (66 records):**
  Characterized by very high discounts, large-value transactions, and concentration in Riyadh. This cluster reflects special cases such as promotions or negotiated deals.

- **Cluster 2 (3,338 records):**
  Contains transactions with lower values across categories, indicating small and regular sales, mainly from the Northern region. This represents low impact, routine customers.

- **Cluster 3 (237 records):**
  Mainly small-value transactions from the Southern region, with minimal medium-category sales. This cluster reflects price-sensitive or low-volume customers.

Overall Insight:

The clustering reveals clear customer segments based on sales value, discount intensity, and location. These segments can support targeted pricing strategies, customer prioritization, and regional sales planning.



Cluster Comparison and Interpretation

The cluster visualizations compare four customer groups across key attributes such as location, value category, quantity, sales amount, and discount using normalized values. Each line represents a cluster's average behavior relative to the overall dataset.

- **Cluster 0 (largest group)** shows balanced behavior across most variables, with slightly higher association with large-value products and the northern region. This cluster represents the typical or average customer segment.
- **Cluster 1 (smallest cluster)** is strongly characterized by extremely high discounts and higher association with large-value products and the Riyadh region. This group likely represents exceptional or promotional transactions.
- **Cluster 2** reflects low values across most dimensions, especially for large and small value categories, indicating low-impact or low-activity customers.
- **Cluster 3** is dominated by small-value products and higher activity in the southern region, with noticeably higher quantities and sales amounts compared to other clusters.

Key Insight

The clustering results reveal clear differences in customer behavior, allowing the business to distinguish between high-discount customers, high-volume small-value buyers, typical customers, and low-impact segments. These insights support targeted pricing strategies, regional planning, and customer prioritization.

Cluster	Location = ال...	Location = الر...	Location = ال...	Location = ال...	Location = الغ...	Value Categ...	Value Categ...	Value Categ...	Quauntity	Amount	Discount
Cluster 0	0.552	-0.159	-0.124	-0.164	-0.122	0.811	-0.726	-0.189	0.060	0.087	-0.031
Cluster 1	-0.480	1.078	-0.154	-0.481	-0.092	1.326	-1.232	-0.189	5.903	6.360	1.246
Cluster 2	-0.517	0.155	0.142	0.104	0.137	-0.754	0.812	-0.189	-0.188	-0.213	-0.047
Cluster 3	-0.011	-0.409	-0.260	0.896	-0.242	-0.754	-1.232	5.283	-0.469	-0.507	-0.037

This table is the **cluster centroid (profile) table** from your clustering model.

What it shows

- **Rows (Cluster 0–3):** Each row represents one cluster.
- **Columns (Location, Value Category, Quantity, Amount, Discount):** These are the variables used to create the clusters.
- **Values (positive/negative numbers):** These **are standardized values (z-scores)** that show how each cluster differs from the overall average.

How to read it

- **Positive value (> 0):** This feature is **higher than average** for that cluster.
- **Negative value (< 0):** This feature is **lower than average** for that cluster.
- **Near 0:** Close to the overall average.

Examples from your table

- **Cluster 1:** Very high Quantity (5.903) and Amount (6.360) → represents high-volume, high-value transactions, often with higher discounts.
- **Cluster 3:** Very high Value Category = Small (5.283) → dominated by small-value category transactions, linked to a specific location.
- **Cluster 0:** Higher Value Category = Large (0.811) → customers mainly buying large-value products with average quantity and amount.
- **Cluster 2:** Mostly negative values → represents low-activity or average customers with smaller quantities and amounts.

Why it is important

This table explains **why each cluster is different**. It helps you:

- Understand customer or transaction behavior in each segment
- Label clusters (e.g., high-value, discount-driven, low-activity)
- Use clusters for targeted strategies in sales, pricing, or credit control

How Clustering Helps the Business

Clustering groups similar sales transactions together based on location, product value category, quantity, amount, and discount. This helps the business **understand different customer and sales behaviors**, rather than treating all sales as one group. With clustering, management can design **targeted pricing, promotion, and inventory strategies** for each segment, improving revenue and efficiency.

Cluster Profiles and Business Strategies

Cluster 0 – High-Value, Stable Sales (Largest Group)

- Dominated by Large value products
- Moderate quantity and sales amount
- Low discounts
- Strong presence in specific regions

Strategy:

- Focus on retention and availability of high-value products
- Maintain pricing with minimal discounts
- Ensure consistent inventory levels

Cluster 1 – High Volume & High Discount Sales (Small but High Impact)

- Very high quantity and sales amount
- Extremely high discounts
- Strong association with large value products
- Concentrated in a specific location

Strategy:

- Use controlled promotions and discount limits
- Analyze profitability to avoid margin loss
- Target during campaigns or seasonal demand

Cluster 2 – Low-Value, Routine Transactions

- Lower quantity and sales amount
- Minimal discounts
- Mainly Medium or mixed value products
- Represents regular, low-risk transactions

Strategy:

- Improve efficiency through bundles or cross-selling
- Automate sales processes
- Use light promotions to increase basket size

Cluster 3 – Small-Value, Location-Specific Buyers

- Strong preference for Small value products
- Lower sales and quantity
- Concentrated in a specific region

Strategy:

- Focus on **volume growth**
- Offer targeted regional promotions
- Introduce upselling to Medium products

Overall Insight

The clustering reveals that sales performance varies strongly by product value, discount behavior, and location. By tailoring strategies to each cluster, the business can increase profitability, optimize promotions, and improve customer targeting, instead of using a one-size-fits-all approach.

Tools Research and Selection Effort

Several analytical and business intelligence tools were reviewed and utilized throughout this project to support data analysis, modeling, and visualization tasks.

Microsoft Excel was initially used for data inspection, cleaning, descriptive statistics, pivot tables, and basic visualizations. Its simplicity and flexibility made it suitable for early-stage data understanding and exploratory analysis.

Power BI was chosen for dashboard development and final presentation. It enabled the creation of interactive dashboards, KPIs, slicers, and visual insights that support managerial decision-making. Power BI's ability to integrate multiple datasets and present results in a clear, business-oriented format made it ideal for communicating analytical findings.

RapidMiner was selected as the primary machine learning platform due to its user-friendly visual workflow and powerful built-in algorithms. It was used to develop predictive models (Linear Regression and Random Forest) for sales prediction and to perform K-Means clustering for customer segmentation. RapidMiner also provided performance metrics that helped evaluate model accuracy and reliability.

Together, these tools formed a complete analytics pipeline—from data preparation and exploration to predictive modeling and interactive visualization—ensuring accurate analysis and actionable business insights.

Project Deployment Effort – Use Case

This project provides an integrated analytical solution that enables the business to monitor performance, predict future outcomes, and segment customers for targeted decision-making.

1. Power BI Dashboard – Monitoring & Decision Support

The interactive Power BI dashboards allow business users (management, sales, and finance teams) to easily monitor sales and collections performance in real time.

- Managers can track key KPIs such as total sales, quantities, discounts, and number of transactions.
- Sales performance can be analyzed by salesperson, customer, product category, location, and time period.
- Trends, imbalances, and high-performing or underperforming areas are quickly identified.

This helps the business make faster, data-driven operational decisions, such as reallocating sales resources, focusing on top locations, or adjusting product strategies.

2. Prediction Model – Forecasting & Planning

The prediction models (Linear Regression and Random Forest) are used to estimate future sales amounts based on historical transaction data.

- The business can forecast expected sales under different conditions (quantity, discounts, customer behavior).
- Management can use these predictions for sales planning, budgeting, inventory control, and target setting.
- The Random Forest model provides more accurate predictions, while Linear Regression offers interpretability and understanding of key drivers.

This enables the business to move from reactive reporting to proactive planning.

3. Clustering – Customer Segmentation & Strategy

Clustering analysis groups customers with similar sales and purchasing behaviors.

- High-value customers can be prioritized for retention and loyalty programs.
- Medium-value customers can be targeted with upselling and promotional strategies.
- Low-value or irregular customers can be managed with cost-efficient sales approaches.

By understanding each customer group, the business can design **targeted sales, pricing, and collection strategies**, improve profitability and reducing risk.

Overall Business Value

By combining dashboards, prediction models, and clustering:

- The dashboard answers “What is happening now?”
- The prediction model answers, “What is likely to happen next?”
- Clustering answers “Who should we focus on and how?”

Together, these tools support smarter sales planning, better customer management, improved financial performance, and stronger strategic decision-making.

Results

The results of this project demonstrate that combining descriptive analytics, predictive modeling, and clustering provides strong and actionable insights into sales performance and customer behavior. Through descriptive statistics and Power BI dashboards, it was observed that sales and quantities are highly right-skewed, where a small number of large transactions and customers contribute disproportionately to total revenue. The most important dashboard insight shows that sales performance is concentrated among specific months, salesmen, product categories, and locations. In particular, a limited group of salesmen and high-value product categories drive most of the total sales, while many transactions remain small in value. This highlights a dependency on key performers and products, which represents both an opportunity and a potential business risk.

The predictive modeling results further strengthened the analysis. Linear Regression provided a clear understanding of how key variables such as quantity and discounts influence sales amount, while Random Forest achieved stronger predictive accuracy by capturing non-linear relationships. High correlation and R^2 values indicated that the models successfully explained a large portion of sales variability, while acceptable RMSE values confirmed practical prediction reliability. These results show that the business can reasonably forecast future sales amounts and use predictions for planning, budgeting, and inventory decisions.

Clustering analysis produced meaningful customer segments based on sales behavior and transaction patterns. The clusters revealed clear differences between high-value, medium-value, and low-value customer groups. This segmentation enables the business to prioritize high-value customers for retention, target growth opportunities among medium-value customers, and manage low-value customers more efficiently. Overall, the most important insight from this project is that sales performance is highly concentrated and predictable when key drivers are considered. By integrating dashboards, prediction models, and clustering, the business gains a comprehensive decision-support system that improves strategic planning, customer management, and overall financial performance.